



JERUSALEM COLLEGE OF ENGINEERING

AN AUTONOMOUS INSTITUTION

Approved by AICTE, New Delhi, Affiliated to Anna University Chennai,
Accredited by NBA, New Delhi and Accredited by NAAC with "A" Grade.
Velachery Main Road, Narayanapuram, Pallikaranai, Chennai, Tamil Nadu - 600100.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

QUESTION BANK

Subject Code : JAL2007

Subject Name: Storage Technologies

Year / Semester : III / V

Regulation : 2023

Branch : CSE (AI&ML)

Unit No	Unit Name	Prepared by	
		Name	Department
I	STORAGE SYSTEMS	Mrs.Haritha J, Assistant Professor	AI&ML
II	INTELLIGENT STORAGE SYSTEMS AND RAID		
III	STORAGE NETWORKING TECHNOLOGIES AND VIRTUALIZATION		
IV	BACKUP - ARCHIVE AND REPLICATION		
V	SECURING STORAGE INFRASTRUCTURE		

Course Instructor / Course Coordinator

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Designation: Professor
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Regulation	:	2023
Course Name	:	STORAGE TECHNOLOGIES
Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No.	:	I
Unit Title	:	STORAGE SYSTEMS
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART – A

Q. No.	Questions	Marks	BTL	CO
1.	Differentiate between structured and unstructured data with suitable examples.	2	K2	CO1
2.	Highlight the role of information storage in supporting business operations.	2	K2	CO1
3.	List the importance of scalability and availability as key characteristics of a data center.	2	K1	CO1
4.	Interpret the impact of evolving computing platforms on information storage requirements.	2	K2	CO1
5.	Outline the concept of Information Lifecycle Management and its role in managing organizational data.	2	K2	CO1
6.	Compare cloud computing and traditional computing infrastructure.	2	K2	CO1
7.	State the significance of resource pooling in cloud computing.	2	K1	CO1
8.	Differentiate the major cloud deployment models based on infrastructure ownership, control, and accessibility.	2	K2	CO1
9.	State the contributions of social networking and mobile computing to third platform technologies.	2	K1	CO1
10.	Illustrate the role of compute virtualization and software-defined data centers in improving data center resource utilization.	2	K2	CO1



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PART – B

Q. No.	Questions	Marks	BTL	CO
1.	Explain the different types of digital data and discuss the fundamental concepts of information storage and the key characteristics of a modern data center.	16	K2	CO1
2.	Analyze how the evolution of computing platforms and Information Lifecycle Management can be integrated to manage data effectively throughout its lifecycle across different computing environments.	16	K4	CO1
3.	Apply the essential characteristics of cloud computing to analyze different cloud services and deployment models, and determine their suitability for different organizational requirements with suitable examples.	16	K3	CO1
4.	Explain the role of Big Data analytics, social networking, and mobile computing in supporting third platform technologies with suitable examples.	16	K2	CO1
5.	Apply the characteristics of third platform infrastructure to examine how its key imperatives drive organizations towards third platform transformation.	16	K3	CO1
6.	Analyze the major building blocks of a data center and assess the role of compute systems and compute virtualization. Evaluate how a Software-Defined Data Center enhances flexibility, scalability, and efficient resource management.	16	K4	CO1



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Regulation	:	2023
Course Name	:	STORAGE TECHNOLOGIES
Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No	:	II
Unit Title	:	INTELLIGENT STORAGE SYSTEMS AND RAID
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART – A

Q.No.	Questions	Marks	BTL	CO
1.	State the role of cache memory in improving the performance of an intelligent storage system.	2	K1	CO2
2.	Define the addressing mechanism used in a hard disk drive. Mention the role of Logical Block Addressing (LBA) in identifying data blocks.	2	K1	CO2
3.	How do the major components of an intelligent storage system work together to provide data access?	2	K2	CO2
4.	Differentiate between scale-up and scale-out storage architectures.	2	K2	CO2
5.	State the role of cache memory in improving the performance of a storage system.	2	K1	CO2
6.	List the RAID levels that provide protection against a single disk failure.	2	K1	CO2
7.	Highlight the differences between HDD and SSD in terms of latency and I/O performance.	2	K2	CO2
8.	Outline the characteristics of scale-up and scale-out storage architectures.	2	K2	CO2
9.	Illustrate the role of data striping in improving read and write performance across multiple disks.	2	K2	CO2
10.	An organization needs both improved performance and protection against disk failure. Which RAID level could be selected, and why?	2	K2	CO2



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PART – B

Q. No.	Questions	Marks	BTL	CO
1.	Explain the major components of an intelligent storage system and discuss how these components support efficient data access and management.	16	K2	CO2
2.	Analyze the addressing mechanisms and performance characteristics of hard disk drives and solid-state drives, and evaluate their suitability for different enterprise storage requirements with suitable examples.	16	K4	CO2
3.	Apply the performance parameters of HDDs and SSDs to select suitable storage devices for different application requirements. Justify the selection based on capacity parameters	16	K3	CO2
4.	Elaborate on the different RAID levels and discuss their working principles, features, advantages, and limitations with suitable examples.	16	K2	CO2
5.	Apply the concepts of different intelligent storage systems to demonstrate how their features and benefits can support various applications in enterprise environments.	16	K3	CO2
6.	Analyze scale-up and scale-out storage architectures based on scalability, performance and resource management. Evaluate their suitability for different storage requirements with appropriate examples.	16	K4	CO2



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Regulation	:	2023
Course Name	:	STORAGE TECHNOLOGIES
Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No	:	III
Unit Title	:	STORAGE NETWORKING TECHNOLOGIES AND VIRTUALIZATION
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART – A

Q. No.	Questions	Marks	BTL	CO
1.	Differentiate between block-based and file-based storage systems.	2	K2	CO3
2.	List the role of object-based storage in managing large volumes of unstructured data.	2	K1	CO3
3.	State the purpose of Converged Enhanced Ethernet (CEE) in an FCoE SAN	2	K1	CO3
4.	Summarize the role of VLAN in an iSCSI-based storage network.	2	K2	CO3
5.	Highlight the characteristics of block-based storage and its method of accessing data.	2	K2	CO3
6.	Outline the role of zoning in restricting communication between servers and storage devices in an FC SAN.	2	K2	CO3
7.	A storage network requires multiple physical links to operate as a logical connection and provide higher bandwidth. Which technique can be used?	2	K2	CO3
8.	State the purpose of iSCSI in connecting servers to block-level storage over an Ethernet/IP network.	2	K1	CO3
9.	Outline the importance of Fibre Channel over Ethernet (FCoE) in carrying Ethernet and Fibre Channel traffic over the same physical network infrastructure.	2	K1	CO3
10.	Illustrate the role and characteristics of the FCIP protocol in connecting Fibre Channel SANs across geographically distributed sites using an IP network.	2	K2	CO3



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PART – B

Q. No.	Questions	Marks	BTL	CO
1.	Describe the architecture and major components of a Fibre Channel SAN. Explain the data flow between the server and storage system with a neat diagram.	16	K2	CO3
2.	Analyze the Fibre Channel Arbitrated Loop and Switched Fabric topologies of FC SANs by comparing their architecture, connectivity and suitability for storage environments.	16	K4	CO3
3.	Apply link aggregation, zoning, and FC SAN virtualization to improve the availability and security of a FC SAN. Illustrate the working of each technique with suitable examples.	16	K3	CO3
4.	Interpret the iSCSI protocol and discuss the network components and connectivity methods used in an iSCSI SAN with a neat diagram.	16	K2	CO3
5.	Apply link aggregation, switch aggregation, and VLAN concepts to demonstrate how bandwidth, redundancy, traffic isolation, and performance can be improved in an IP-based SAN.	16	K3	CO3
6.	Analyze the Fibre Channel over Ethernet (FCoE) SAN architecture, including its components, connectivity, and data flow. Evaluate the advantages and limitations of FCoE SAN in data center environments.	16	K4	CO3



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Regulation	:	2023
Course Name	:	STORAGE TECHNOLOGIES
Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No	:	IV
Unit Title	:	BACKUP - ARCHIVE AND REPLICATION
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q. No.	Questions	Marks	BTL	CO
1.	List the role of business continuity in an organization	2	K1	CO4
2.	Highlight the major components of a backup architecture.	2	K2	CO4
3.	Outline the role of replication in maintaining data availability.	2	K2	CO4
4.	Differentiate between storage-based replication and network-based replication.	2	K2	CO4
5.	State the backup approaches used for regular protection of critical data.	2	K1	CO4
6.	Define the purpose of data deduplication in reducing backup storage capacity.	2	K1	CO4
7.	List the backup approaches used to protect data stored on smartphones and tablets.	2	K1	CO4
8.	An organization needs to retain old business records that are rarely accessed but must be preserved for several years. Should backup or archive be preferred? Explain.	2	K2	CO4
9.	A company needs to transfer its existing data from an old storage system to a new storage platform without losing the data. Which process should be used?	2	K2	CO4
10.	Outline the role and key features of Disaster Recovery as a Service (DRaaS) in providing cloud-based disaster recovery capabilities.	2	K2	CO4



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PART B

Q. No.	Questions	Marks	BTL	CO
1.	Discuss the concept of Business Continuity, including its need and major components in an organizational strategy. Explain the role of backup and recovery in maintaining continuous business operations.	16	K2	CO4
2.	Analyze the backup architecture with a neat diagram and compare different backup targets and methods based on their operation, performance, and suitability for various backup requirements.	16	K4	CO4
3.	Apply data deduplication techniques in backup environments and their working principles. Illustrate how their advantages and limitations affect storage utilization and backup performance.	16	K3	CO4
4.	Describe the cloud-based backup and mobile device backup in detail. Discuss their architecture, working and suitable use cases in an enterprise environment.	16	K2	CO4
5.	Apply the concepts of compute-based, storage-based, and network-based replication to demonstrate their working mechanisms, advantages, limitations, and suitability for different applications.	16	K3	CO4
6.	Analyze the role of data migration and Disaster Recovery as a Service (DRaaS) in improving disaster recovery. Explain the DRaaS architecture, working and suitable data migration strategies.	16	K4	CO4



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Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No	:	V
Unit Title	:	SECURING STORAGE INFRASTRUCTURE
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q. No.	Questions	Marks	BTL	CO
1.	List the importance of confidentiality, integrity, and availability in storage security.	2	K1	CO5
2.	State the role of storage security domains in protecting a storage infrastructure.	2	K1	CO5
3.	Highlight the purpose of security controls in a storage environment.	2	K2	CO5
4.	Outline the importance of monitoring and reporting in storage infrastructure management	2	K2	CO5
5.	State the security controls used to protect confidential customer information in a storage system.	2	K1	CO5
6.	Identify the security mechanism used to restrict employee access to storage resources based on job roles.	2	K1	CO5
7.	Name the risk management activity used to identify potential security problems in a storage infrastructure.	2	K1	CO5
8.	Define how the compliance activity in GRC helps an organization ensure that its storage infrastructure follows security policies and applicable regulations.	2	K2	CO5
9.	How capacity management can be used to address frequent capacity limitations in a storage system.?	2	K2	CO5
10.	Outline how performance management can be applied to identify and resolve repeated performance degradation in a storage system.	2	K2	CO5



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PART B

Q. No.	Questions	Marks	BTL	CO
1.	Explain the goals of information security and discuss how confidentiality, integrity, and availability can be maintained in a storage infrastructure with suitable examples.	16	K2	CO5
2.	Analyze the major storage security domains and examine the common threats associated with each domain to determine their impact on protecting a storage infrastructure.	16	K4	CO5
3.	Apply suitable security controls to protect a storage infrastructure against common security threats. Explain how these controls help maintain data confidentiality, integrity, and availability.	16	K3	CO5
4.	Apply the concepts of Governance, Risk, and Compliance (GRC) to demonstrate how they support the management of a secure and compliant storage infrastructure.	16	K3	CO5
5.	Discuss the major storage infrastructure management functions and processes, highlighting their role in efficient storage operations.	16	K2	CO5
6.	Analyze the storage infrastructure management functions and processes, and evaluate their effectiveness in improving capacity, performance, availability, and overall storage efficiency.	16	K4	CO5